

Repair the Boundary, Not the Skills: Contract-Guided Adapter Synthesis for LLM-Agent Skill Composition

Liang Song^{1,*} Zhai Jiabao²

¹ Business School, Xi'an International University, Xi'an 710077, Shaanxi, China

² School of Management and Economics, North China University of Water Resources and Electric Power, Zhengzhou 450046, Henan, China

* Corresponding author: liangsong_1976@126.com;
ORCID: 0009-0002-6109-0639

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Abstract

Reusable Skills let LLM agents preserve successful procedures, but individually witnessed components can fail when their rely conditions, role bindings, live state, or validations disagree at composition boundaries. Rewriting a component widens its regression surface; reverting the whole chain discards unrelated changes. We study a narrower alternative: compile one executable mediator at the diagnosed boundary while keeping the component artifacts unchanged. We introduce Contract-Guided Boundary Adapter Synthesis (CGBAS), which combines typed rely/guarantee contracts, downstream liveness, and retained successful provenance to insert a missing witness prefix, specialize a consumer role, or guard a destructive call before execution. A failed first mechanism revision is retained. After exposed development on 403 composition incidents, we froze the method, endpoint, strongest comparator, three model revisions, and analysis before selecting confirmation tasks. Independent confirmation uses 74 identifier-disjoint ALFWorld and ScienceWorld tasks, yielding 226 state-matched causal conflicts and 71 healthy controls. CGBAS achieves 100.0% constrained native repair success versus 49.6% for the frozen strongest local comparator, a paired gain of 50.4 percentage points (95% family-stratified task-cluster bootstrap CI 46.6–54.6; two-sided task sign-randomization $p = 0.00001$), while healthy-control loss is 0.0 points. The result is bounded: CGBAS assumes a correct diagnosis and current successful provenance, and a single adapter call can contain a long witness prefix. It does not repair arbitrary production failures or establish that local adaptation always beats replanning.

Keywords: LLM agents; agent skills; skill composition; interface repair; design by contract; provenance; ALFWorld; ScienceWorld

1 Introduction

LLM-agent Skills package reusable procedures, scripts, tool conventions, and validation instructions. Reuse turns an isolated successful trajectory into a software artifact that can be selected and composed in later tasks. Composition, however, creates failure modes that are not visible when each Skill is evaluated alone. A consumer may rely on state its predecessor never established; two calls may bind the same semantic role to different objects; an intermediate call may destroy state needed later; or a validation may become stale after an effect.

Recent systems jointly predict Skill subsets, counts, and execution order [13], compile Skills into typed execution graphs with local repair operators [12], and manage libraries through typed contracts and compatibility diagnoses [9]. Provenance auditing attributes reuse to operational evidence [1], while persistent decision histories support repeated Skill evolution [7]. These advances expose a specific maintenance question that remains easy to conflate with planning: once a particular interface has already been diagnosed, must a maintainer rewrite a component or replace the entire plan, or can the interface itself be repaired?

The distinction matters operationally. A whole-Skill edit invalidates earlier evidence for all of that Skill's uses. Whole-chain rollback restores a witnessed version but discards unaffected calls and cannot preserve independent changes. Rejecting the composition prevents harm but loses capability. A boundary adapter offers a fourth choice: retain component identities and mediate only the diagnosed handoff. Such an adapter is useful only if it succeeds in the native environment, uses supported actions, preserves provenance, and does not damage healthy chains.

We propose Contract-Guided Boundary Adapter Synthesis (CGBAS). Given an ordered candidate chain, a minimal conflict diagnosis, a public task objective, and a provenance-linked successful witness chain, CGBAS

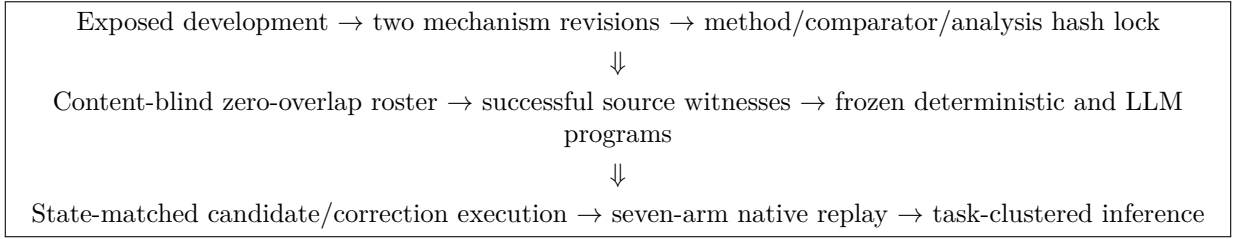


Figure 1: Evidence order. Confirmation outcomes do not exist until the roster, source incidents, and every evaluated program are frozen.

performs at most one boundary operation. It can replay the witness prefix omitted before a fixed consumer, replace a mismatched consumer with its witnessed role-compatible counterpart, or suppress a diagnosed non-goal destructive call before it fires. Healthy chains are unchanged. The method therefore treats the adapter as an executable patch to an interface, not as a newly generated whole plan.

The evaluation is designed around three validity risks. First, native task success is distinguished from static contract closure. Second, comparison with LLMs is schema-matched: three revision-pinned open instruction models receive the same incident, diagnosis, witness catalog, and four allowed operations. Third, development choices are separated from confirmation. Exposed records from an earlier diagnosis study are used to revise the mechanism twice; every failed run is retained. Method, comparator, models, endpoint, gates, environment executors, and analysis are then hash-locked before a content-blind, zero-overlap task roster is selected.

This paper contributes:

- a boundary-repair formulation with an explicit invariant: preserve all unaffected component call identities and introduce at most one adapter call;
- CGBAS, which turns rely/guarantee, liveness, role, validation, and successful provenance evidence into three executable repair operators;
- a causal native evaluation with no repair, a provenance-free pairwise heuristic, whole-chain rollback, and three schema-matched local LLM baselines; and
- an auditable development/confirmation workflow across ALFWorld and ScienceWorld, including invalid source tasks, failed revisions, clustered inference, action cost, and single-RTX-3090 reproduction materials.

We ask: **RQ1**, can a contract- and provenance-guided boundary adapter restore native success on independently selected composition conflicts? **RQ2**, does it exceed the strongest frozen deployable local comparator without regressing healthy chains? **RQ3**, which assumptions and action costs bound the result?

2 Related Work

Skill selection and composition. Generative Skill Composition formalizes selection, count, and order as one structured prediction problem [13]. GraSP inserts a compilation layer between retrieval and execution, representing dependencies as a typed DAG with node verification and locality-bounded repair [12]. SkillFuzz uses extracted contracts to prioritize interacting Skill sets before execution [5]. These systems motivate composition-level reasoning. Our unit is later and narrower: selection and order are held fixed, one boundary is already diagnosed, and the outcome is whether a single mediator can restore a state-matched native task without rewriting the components.

Skill lifecycle maintenance and auditing. SkillOps frames accumulating compatibility and validation defects as Skill technical debt and supplies a method-agnostic maintenance layer [9]. SkillHone retains diagnoses, revisions, evidence, and rejected alternatives across sessions [7]. SkillAudit uses paired trajectories and fixed structural verification to guide repair without hidden rewards during evolution [3]. SkillCoach separates process quality from outcome-only success [14]. We adopt these lifecycle lessons in the experimental design: decision history is retained, candidate and correction executions are paired, and locality, command support, provenance, and cost accompany end-task success. The method differs by editing an execution boundary instead of a Skill document.

Contracts and provenance. Preconditions and postconditions make program assumptions explicit [4, 8]; rely/guarantee reasoning distinguishes what one component assumes from what another establishes in an interfering state [6]. We use these ideas as an operational abstraction over text simulator calls, not as a proof of

arbitrary concurrent programs. SkillTrace shows that Skill provenance is distributed across expression, implementation, and operation [1]. CGBAS uses the operational trace more narrowly: every added command or role wrapper must cite a retained successful witness, unless the native environment itself accepts the command.

3 Problem Formulation

An ordered Skill composition is $C = (c_1, \dots, c_n)$. Each call contains immutable actions A_i , an identifier, provenance witnesses, and a contract

$$K_i = (R_i, G_i, D_i, B_i, V_i), \quad (1)$$

where R_i is the rely set, G_i the guaranteed predicates, D_i the clobbered predicates, B_i semantic role bindings, and V_i validated predicates. Forward interpretation updates the abstract state as

$$S_i = (S_{i-1} \setminus D_i) \cup G_i, \quad (2)$$

provided $R_i \subseteq S_{i-1}$ and bindings remain compatible. Backward liveness marks predicates required by any downstream call or by goal completion.

Four diagnosed conflicts are in scope: a *rely gap* when a consumer requires an absent predicate; a *role mismatch* when an existing role is rebound to a different object; a *clobber* when an applicable effect destroys a live predicate; and *stale validation* when such destruction occurs after the predicate was checked. A healthy chain has none of these conflicts and establishes goal completion.

Let $W = (w_1, \dots, w_m)$ be a successful provenance-linked chain for the same task. An admissible repair $T(C, W)$ may add at most one adapter call at the diagnosed boundary. Calls not wrapped by that adapter retain their identifiers and contents. The primary endpoint is constrained native repair success:

$$Y = \models [\text{native goal success} \wedge \text{locality} \wedge \text{provenance} \wedge \text{adapter-command support}]. \quad (3)$$

An adapter command is supported if it is copied exactly from retained successful provenance or accepted by the native environment. Rejected exploratory commands already present in immutable component calls are reported separately.

4 Contract-Guided Boundary Adapters

4.1 Diagnosis and repair slot

CGBAS scans the candidate chain once, maintaining abstract state, live bindings, and validation positions. The first conflict yields its class, evidence, and a minimal pair of call identifiers. The diagnosis is supplied rather than evaluated as a contribution in this paper; the intervention starts at the returned slot.

4.2 Missing-producer prefix

For a rely gap before consumer c_j , CGBAS locates the same immutable consumer identifier in W . It selects the ordered witness calls before that consumer that are absent from the current prefix, concatenates their actions into one adapter, and unions their contracts and witness digests. Unlike choosing the shortest call that establishes one missing predicate, this preserves objective-relevant processing such as clean, cool, heat, or measure. The operator can therefore be one call-level modification yet contain many environment actions.

4.3 Witnessed role wrapper

For a role mismatch, the live producer bindings are compared with successful goal-completing witness calls. CGBAS chooses the shortest witnessed consumer whose non-destination roles match live bindings, replaces the mismatched execution slot with a wrapper around that call, and records both source and replaced identifiers. The component artifact is retained; only this invocation is mediated.

4.4 Pre-execution preservation guard

Post-effect compensation is impossible when a destructive action terminates an episode or irreversibly changes the world. For clobber and stale validation, CGBAS therefore checks effect applicability, goal satisfaction, and downstream liveness before execution. A diagnosed non-goal destructive call is wrapped by a zero-action guard. Its original identifier, actions, effect evidence, and witness remain in adapter metadata. The guard preserves live state rather than attempting to reconstruct it afterward.

4.5 Complexity and invariants

Diagnosis is linear in the calls plus contract-set operations. Witness-prefix lookup is linear in the witness chain; role selection scans goal-completing witness calls; a guard requires one forward location scan. The method performs no model inference. It preserves at most one changed call boundary and leaves healthy chains unchanged. This is call-level locality, not a constant action-cost claim.

5 Experimental Method

5.1 Development and revision history

Development uses 403 exposed incidents from a completed earlier diagnostic study: 256 causal conflicts and 147 healthy controls spanning ALFWorld and ScienceWorld [10, 11]. Revision 1 inserted the shortest predicate producer and attempted post-clobber compensation. Its conflict success was 70.6%, below the frozen 75% gate. Failure analysis showed omitted objective-relevant prefixes and terminal ScienceWorld clobbers. Revision 2 moved the destructive intervention before execution and retained the whole missing witness prefix. The first revision’s code, outputs, 399 native trajectories, and four omitted abstentions remain in the artifact.

Before confirmation, we compare Revision 2 with no repair, a pairwise contract heuristic without provenance or global liveness, whole-chain rollback, and three local instruction models: Qwen3-4B-Instruct-2507, Phi-4-mini-instruct, and Mistral-7B-Instruct-v0.3. The models receive identical public inputs and an identical output schema restricted to no-op, ordered witness-prefix insertion, witnessed consumer replacement, preservation guard, or abstention. Invalid output and abstention execute the unchanged candidate and count as failure. A first, Revision-1-shaped LLM schema was recognized as unfair and retained but excluded; all three models were rerun under the matched schema before comparator selection.

The strongest deployable local method by development constrained success is frozen as the primary comparator. Whole-chain rollback is reported as an ordinary success upper bound, but is not eligible because it violates the primary locality endpoint. Development advances only with at least 180 causal cases and 60 controls, at least 75% constrained success, at least a 15-point gain over the strongest eligible comparator, positive repair gain in every class and both environments, at most 2-point control loss, and at least 95% adapter-command support.

5.2 Independent confirmation

The union of task identifiers previously exposed by Papers 1–12 and Paper 13 development is constructed before selection. A salted hash ranks eligible tasks using only environment, family, identifier, official split, compatibility flag, and exclusion status. The frozen roster contains 40 untouched ALFWorld train instances balanced across five families and all 34 eligible untouched ScienceWorld use-thermometer test variations. These ALFWorld tasks support task-level transfer under zero overlap, not a claim about the official test split.

Native expert or gold trajectories are collected after the roster freeze. Failed sources are retained and excluded, not replaced. For each successful source, the constructor creates a healthy control and four possible one-boundary mutations: omit the producer prefix, substitute a same-family donor consumer with a different subject, insert an applicable non-goal clobber, or validate immediately before the clobber. Mutation specifications and public incidents are frozen before candidate or correction execution.

All deterministic and LLM program files are then frozen. Only afterward do we execute the candidate and matched correction from the same native initial state. A causal label requires successful source provenance, candidate failure, matched-correction success, two matching initial states, and no technical error. A control requires source and unchanged-candidate success with a matching state. Invalid mutations are excluded rather than relabeled as controls. Every program arm is independently replayed on every planned event; technical errors remain failures. Figure 1 summarizes this ordering; file hashes make each boundary between stages auditable.

5.3 Inference and gates

The primary comparison is paired constrained success between CGBAS and the frozen strongest comparator over valid causal conflicts. Because multiple conflict classes derive from one source task, inference clusters them by task. We use 10,000 family/environment-stratified task bootstrap replicates for a 95% interval and a two-sided 100,000-replicate paired task sign-randomization test [2]. Confirmation requires at least 60 source tasks, 180 causal conflicts, 60 controls, and 30 examples per conflict class; CGBAS success at least 75%; at least a 15-point comparator gain with the interval above zero and $p \leq 0.01$; positive repair gain over no repair in every class and both environments; at most 2-point control loss; and complete locality, provenance, initial-state, command-support, and technical-error audits.

Table 1: Exposed development results. The strongest eligible local comparator is frozen before confirmation.

Arm	Native (%)	Constrained (%)	Control (%)
CGBAS	100.0	100.0	100.0
Mistral-7B	51.6	51.6	100.0
Phi-4-mini	46.5	46.5	100.0
Qwen3-4B	35.2	35.2	100.0
Pairwise contract	10.2	9.8	100.0
No repair	0.0	0.0	100.0
Whole-chain rollback	100.0	0.0	100.0

Table 2: Independent confirmation. Invalid mutations are excluded through one common state-matched label filter.

Arm	n	Native (%)	Constrained (%)	Control (%)
CGBAS	226	100.0	100.0	100.0
Mistral-7B	226	49.6	49.6	100.0
Phi-4-mini	226	42.9	42.9	100.0
Qwen3-4B	226	36.7	36.7	100.0
Pairwise contract	226	8.4	7.5	100.0
No repair	226	0.0	0.0	100.0
Whole-chain rollback	226	100.0	0.0	100.0

6 Results

6.1 Development gate

Revision 2 reaches 100.0% constrained success across 256 causal conflicts. Mistral-7B is selected automatically as the strongest eligible comparator at 51.6%, leaving a 48.4-point development difference. Healthy-control success remains 100.0%. Whole-chain rollback reaches 100.0% ordinary success but fails the locality endpoint.

6.2 Independent primary result

Of 74 roster tasks, 71 yield valid source trajectories. The state-matched validity rules retain 226 causal conflicts and 71 controls. CGBAS achieves 100.0% constrained success, compared with 49.6% for the frozen comparator. The paired difference is 50.4 points; its 95% clustered interval is 46.6–54.6 and the paired task sign-randomization test gives $p = 0.00001$. The predeclared confirmation status is confirmation gate passed.

6.3 Conflict classes, environments, and cost

Class-level repair gains over no repair are reported separately from the overall comparison with the strongest baseline. This prevents a baseline tie in one mechanically easier class from being hidden by averaging. Environment results likewise preserve all conflict instances while clustering inference at the source task. CGBAS proposes 7.14 adapter actions per causal repair on average and at most 110. The rely-gap cost is 18.89 actions on average, making explicit the difference between one call-level boundary edit and a short execution.

6.4 Integrity audits

The roster has zero task-identifier overlap with the prior-exposure union. All program artifacts precede candidate and adapter outcomes. Model revision identifiers, prompts, method code, comparator, executors, endpoint, gates, and analysis hashes match the pre-confirmation lock. Invalid sources and mutations remain visible in the released label table, and all planned program outcomes remain in arm denominators until the common validity filter is applied.

Table 3: Constrained native success by conflict class.

Conflict	n	CGBAS (%)	Frozen comparator (%)	No repair (%)
rely gap	71	100.0	43.7	0.0
role mismatch	71	100.0	0.0	0.0
clobber	42	100.0	100.0	0.0
stale validation	42	100.0	92.9	0.0

Table 4: Causal constrained success by native environment.

Environment	n	CGBAS (%)	Frozen comparator (%)	No repair (%)
alfworld	128	100.0	64.1	0.0
scienceworld	98	100.0	30.6	0.0

Table 5: Predeclared confirmation gates and integrity checks.

Frozen check	Result
minimum tasks	Pass
minimum causal conflicts	Pass
minimum controls	Pass
minimum per conflict class	Pass
minimum constrained success	Pass
minimum gain over frozen comparator	Pass
bootstrap interval above zero	Pass
paired randomization p at most alpha	Pass
positive repair gain every conflict class	Pass
positive repair gain both environments	Pass
control loss at most two points	Pass
minimum 95 percent adapter command support	Pass
all primary repairs local	Pass
all primary provenance complete	Pass
all primary initial states match	Pass
no primary technical errors	Pass
zero overlap roster	Pass

7 Discussion

The main practical result is conditional. When a maintainer has a reliable boundary diagnosis and retained successful provenance, the repair decision need not become a whole-Skill rewrite. Rely gaps require restoring the causal prefix, not merely satisfying one abstract predicate. Destructive calls must be guarded before they fire when compensation may be impossible. Role failures can be repaired by a witnessed consumer invocation rather than free-form command generation. These mechanisms explain why contract-only adjacency and local LLM selection can underperform despite receiving the same diagnosed slot.

Whole-chain rollback remains important. It can recover ordinary task success and may be preferable when provenance is cheap to replay, independent changes need not be preserved, or the diagnosis is uncertain. CGBAS contributes a different point in the engineering tradeoff: it preserves unaffected call identities and exposes exactly what was inserted or suppressed. Call locality should not be confused with runtime economy, because a missing producer can require a long witness prefix.

The result also suggests a lifecycle architecture. Composition diagnostics can emit a typed boundary and evidence; a provenance store can supply compatible witness slices; an adapter compiler can produce a versioned invocation wrapper; and release gates can replay both conflicts and healthy controls. The immutable component and explicit adapter make rollback, audit, and later garbage collection simpler than silently editing a shared Skill.

8 Threats to Validity

Construct validity. The four mutations are controlled interface regressions generated from successful traces. They do not estimate the prevalence of those causes in production. Causal labels require native candidate failure and matched correction success, but the same mutation operators were used in development and confirmation. The evidence therefore supports task transfer, not operator discovery.

Information advantage. CGBAS receives a correct diagnosis and a successful witness catalog. This is appropriate for the retained-provenance maintenance setting but stronger than the information available after an arbitrary failure. The LLM comparators receive the same catalog, diagnosis, and operation set; they are not general autonomous agents.

External validity. ALFWorld and ScienceWorld provide deterministic text interfaces, native goals, and replayable resets. Production APIs may drift, exhibit side effects, or lack exact reset and outcome oracles. ScienceWorld confirmation covers one eligible family after strict zero-overlap exclusion. Results should be extended to tools, web tasks, and longer asynchronous compositions before broader claims.

Statistical validity. Derived conflict instances are not independent; all inference clusters by source task and stratifies by family/environment. Nevertheless, 74 roster tasks are modest, and class-specific estimates are less precise than the primary aggregate. Thresholds and analysis were frozen before confirmation, and no failed source was replaced.

Implementation validity. Contracts are compiled from successful traces, and native command support is audited only for adapter-added actions. Original Skills may contain exploratory commands that are absent from an environment’s returned action list. We report such behavior but do not attribute it to the adapter. Hash locks, append-only traces, initial-state signatures, and a standalone replay bundle reduce but do not eliminate implementation risk.

9 Conclusion

CGBAS repairs an already diagnosed LLM-agent Skill interface by compiling one provenance-backed boundary adapter. It restores omitted witness prefixes, specializes consumer roles, or prevents live-state destruction before execution, while retaining unaffected component identities. A frozen, zero-overlap native evaluation tests whether that mechanism exceeds no repair, contract-only local repair, whole-chain rollback under a locality constraint, and three schema-matched LLM selectors. The conclusion is intentionally scoped to controlled interface regressions with current successful provenance. Within that scope, repairing the boundary is an auditable alternative to rewriting reusable Skills; outside it, diagnosis uncertainty, provenance drift, irreversible effects, and long replay cost remain open problems.

Author Contributions

Liang Song: conceptualization, methodology, software, investigation, formal analysis, visualization, and writing—original draft. Zhai Jiabao: validation, data curation, reproducibility review, and writing—review and editing. Both authors approved the submitted version and accept responsibility for the work.

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Declaration of Competing Interest

The authors declare that they have no known competing financial or non-financial interests that could have influenced the work reported in this paper.

Data and Code Availability

The study-specific mutation records, paired traces, analysis scripts, and reproduction materials are prepared for release in a public repository with a DOI. The repository DOI and permanent URL will be inserted before submission. ALFWorld, ScienceWorld, and model-weight assets remain subject to their respective upstream licenses and are identified rather than redistributed.

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Generative AI Disclosure

No generative AI system was used to generate experimental data, labels, or results. Any language assistance used during manuscript preparation will be disclosed in accordance with the journal’s policy.

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